

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I **G. Yashmita (192511032)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

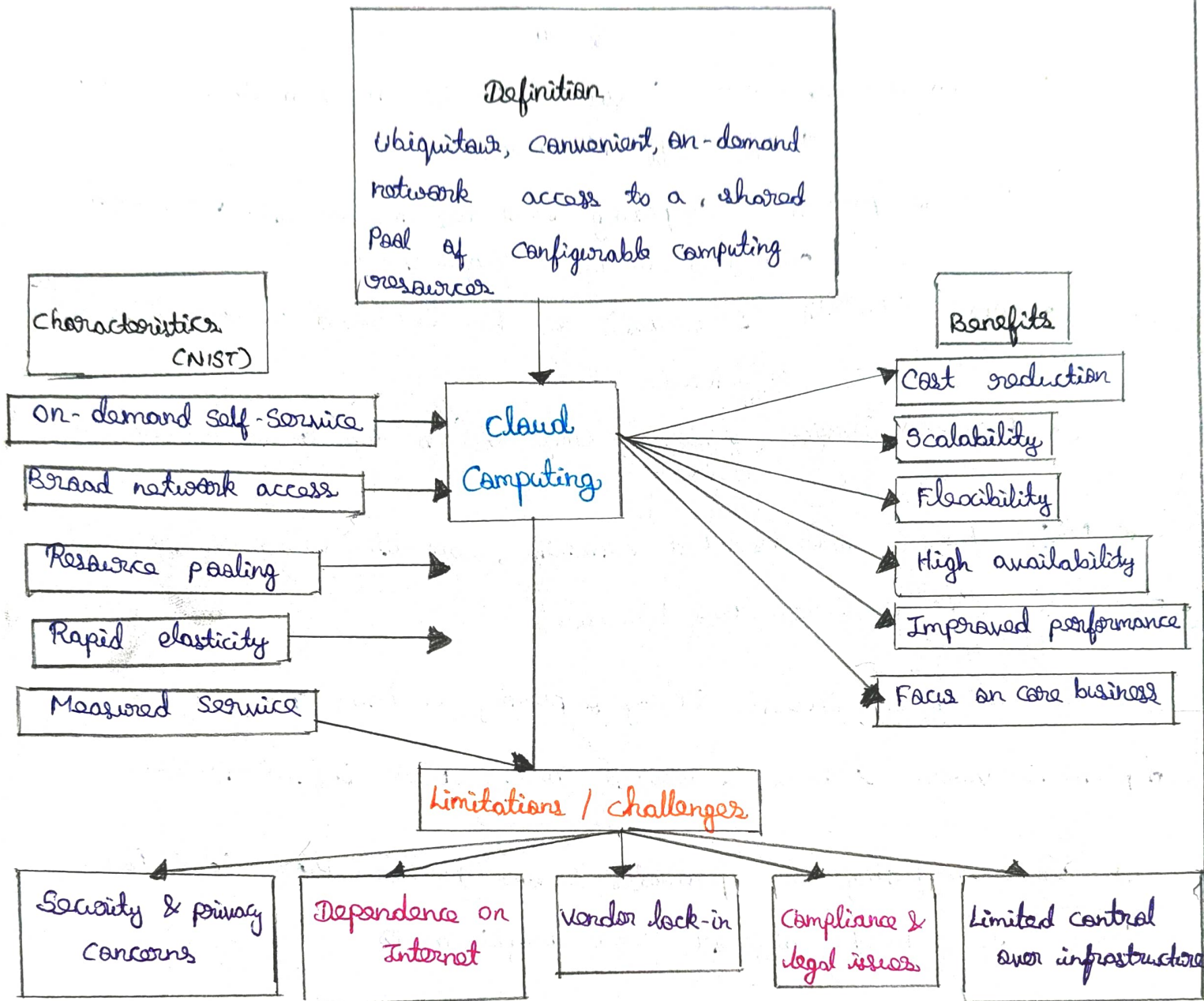
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Concept Map 1: Cloud computing Fundamentals



This concept map explains the fundamentals of cloud computing, including its definition, characteristics, benefits and limitations.

* Cloud computing provides convenient & on-demand access to a shared pool of configurable computing resources through a network.

* The main characteristics are :

- on-demand self-service - resources can be obtained whenever required.
- Broad network access - Services can be accessed through networks using different devices
- Resource pooling - Computing resources are shared & allocated according to user requirements
- Rapid elasticity - resources can be increased or decreased based on demand.
- Measured Service - resource usage can be monitored & measured.

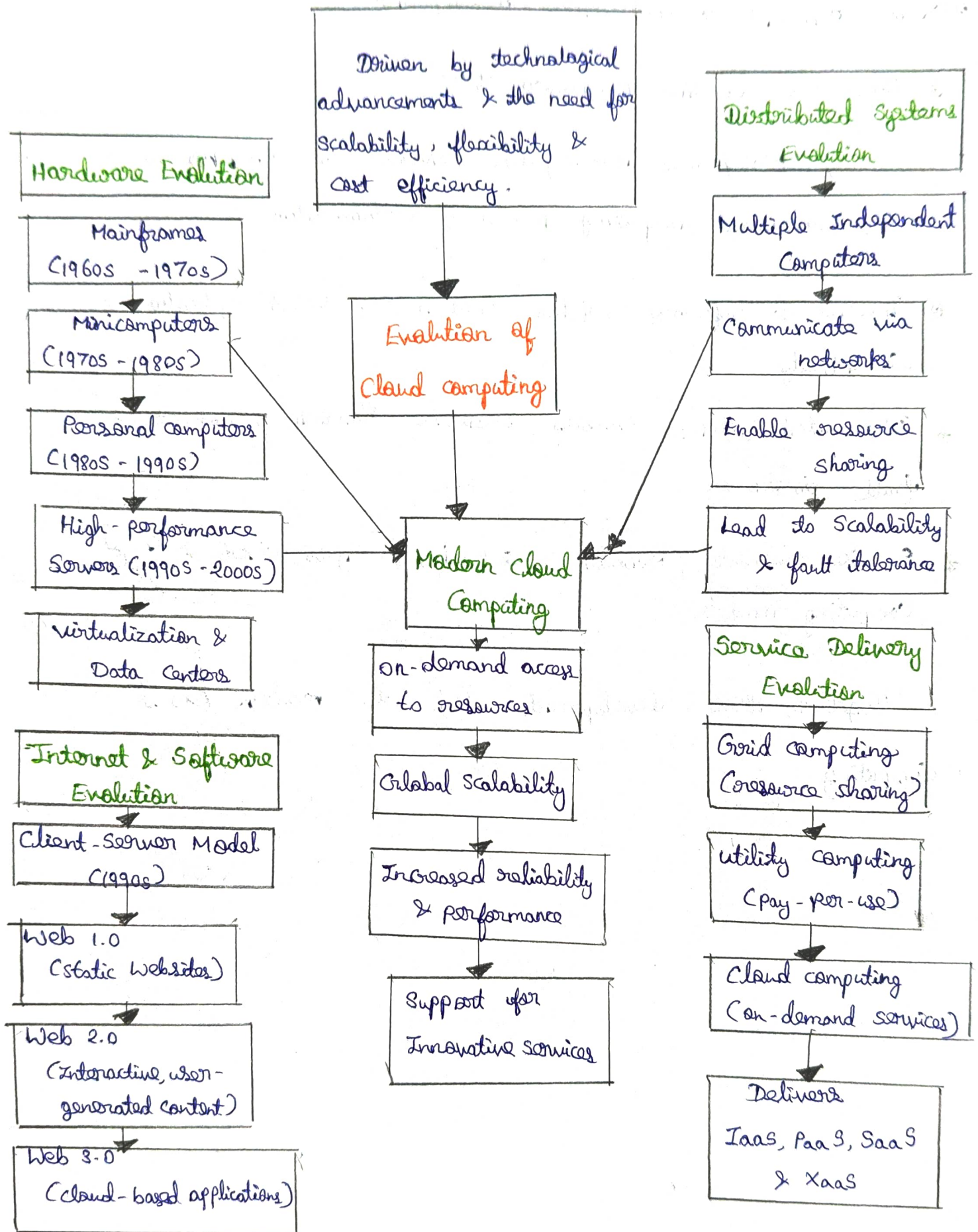
* Major benefits includes cost reduction, scalability, flexibility, high availability & focus on core business.

* The limitation include security & privacy concerns, Internet dependence, vendor lock-in & limited control over infrastructure.

* Cloud computing also provides service models like IaaS, PaaS, SaaS to meet different user requirements

* Cloud services can be deployed as public, private, hybrid or community clouds.

Concept Map 2: Evolution of cloud computing



This concept map explains the evolution of cloud computing through different technological developments.

- * Hardware evolution progressed from:

Mainframes, minicomputers, personal computer and so on.

This increased computing power & virtualization.

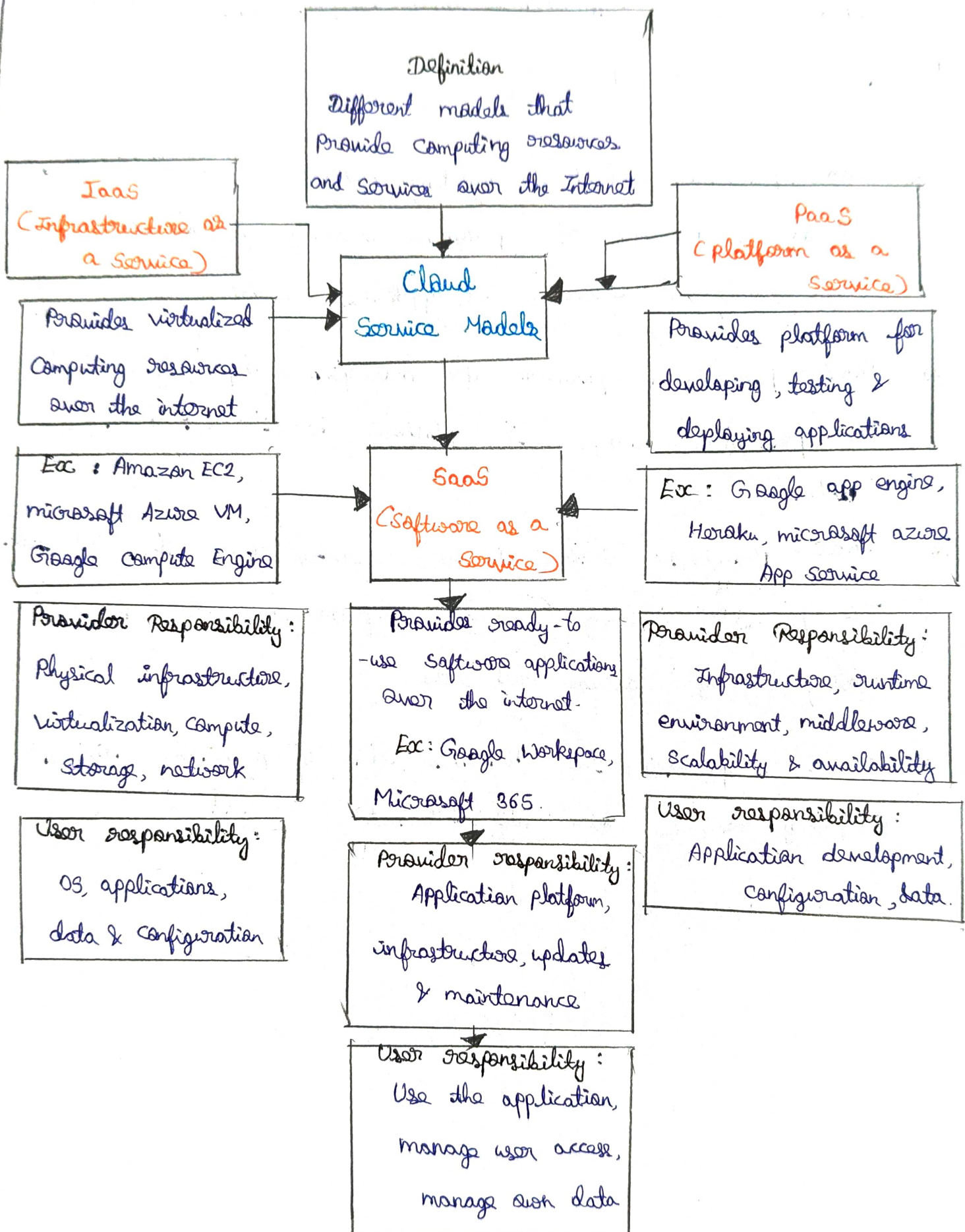
- * Internet & Software evolution Enabled web-based applications & Connectivity.

- * Distributed systems Enabled resource sharing, scalability & fault tolerance.

- * Service delivery evolution introduced grid, utility & cloud computing models.

Together, these developments led to modern Cloud Computing.

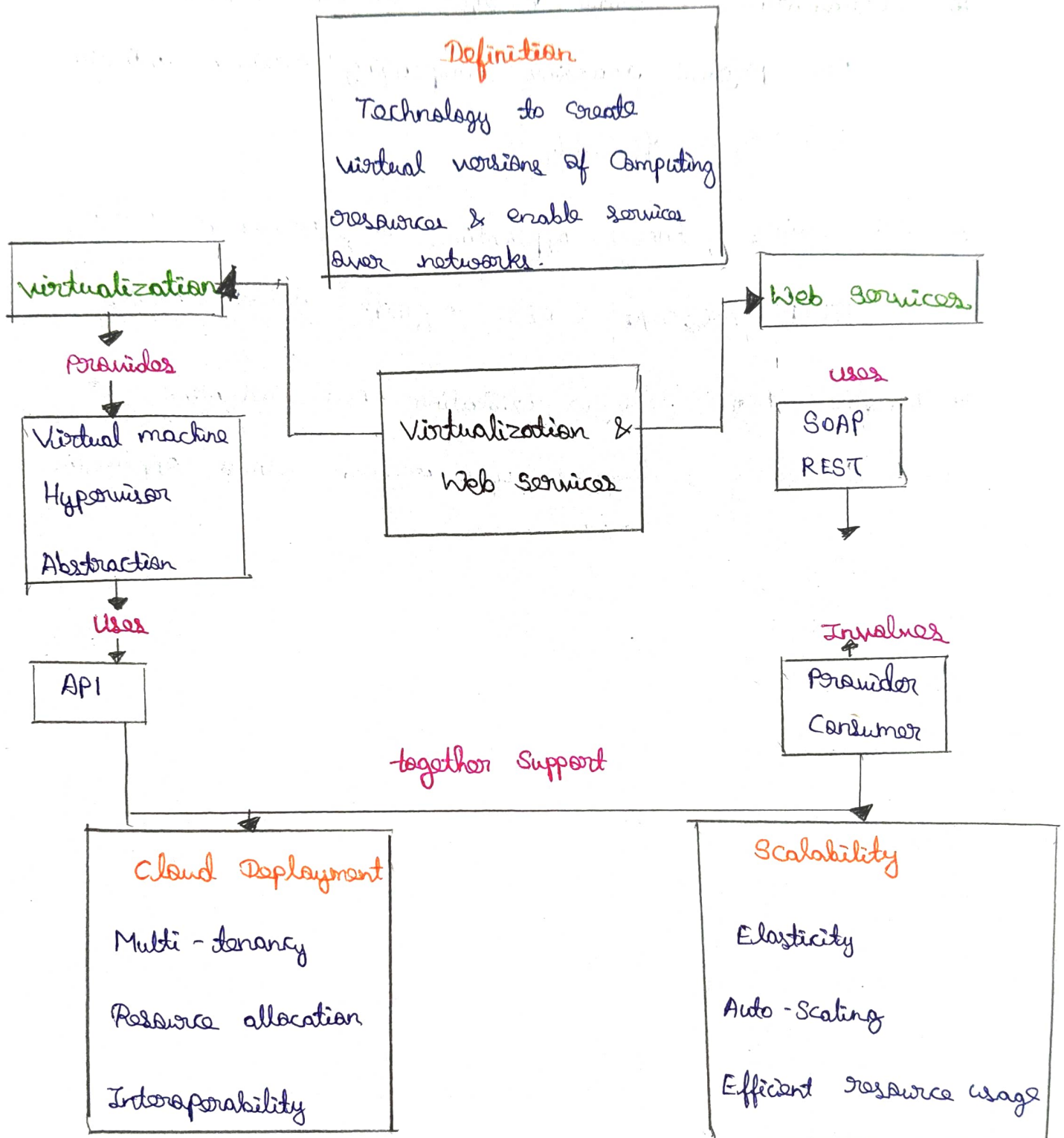
Concept map 3: Cloud Service Models



This Concept map illustrates three main cloud service models - IaaS, PaaS & SaaS - and shows how responsibilities are divided between cloud service provider & user.

- * IaaS: provides virtualized computing resources such as servers, storage & networks. The provider manages infrastructure, while the user manages OS, applications & data.
- * PaaS: provides a platform for developing & deploying applications. The provider manages the platform, while the user manages data.
- * SaaS: provides ready-to-use software over the internet. The provider manages software & infrastructure, while the user mainly manages access and data.
- * XaaS: Represents the broader idea of providing different computing services over the internet.

Concept map 4 : Virtualization & Web Services as Enablers of cloud



This concept map shows how virtualization & web services enable cloud computing.

* Virtualization : Creates multiple virtual machines on one physical machine, improving resource utilization, Scalability & flexibility.

* Web services : Enable applications to communicate using HTTP, XML, SOAP & REST supporting interoperability.

* Benefits : Better resource utilization, cost reduction, Scalability & support for modern cloud applications.